

Własna ocena:

O Artykule

Artykuł opisuje nowatorski wg autorów algorytm przycinania oparty na optymalizacji kolonii mrówek (ACO-P), który dynamicznie kompresuje konwolucyjne sieci neuronowe podczas procesu uczenia.

Plusy

- Efektywność pruningu
- Szybkość obliczeniowa ACO-P
- Dobrze objaśnione parametry CNN training and ACO algorithm (choć autorzy nie piszą skąd taki ich dobór)
- Autorzy piszą o słabościach rozwiązania (co jest jeszcze do zrobienia)

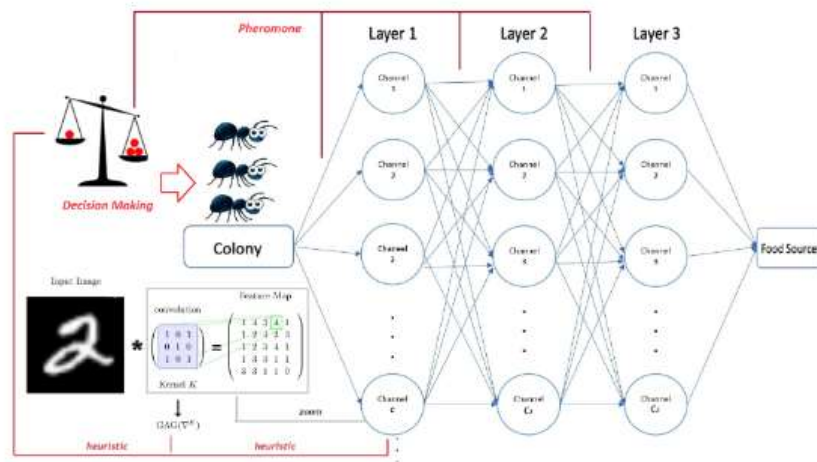
Minusy

- Mało eksperymentów, test tylko na zbiorze MNIST
- Nie wiadomo nic o skalowalności do nowych metod uczenia maszynowego (Porównanie do RNN-P - stara metoda)
- Metoda ACO powoduje spadek accuracy względem modelu bez pruningu - nie jest wytłumaczone co nam ten pruning daje

Approach	Training (AvACC%)	Testing (AvACC%)	Time for Training (s)	Pruning Rate (%)
No pruning	99.60 _{0.001}	99.01 _{0.001}	235.71 _{3.17}	-
Random-P	57.31 _{0.36}	58.18 _{0.37}	271.23 _{1.75}	60 _{0.40}
RNN-P	93.14 _{0.003}	93.42 _{0.008}	683.4 _{20.226}	14 _{0.039}
ACO-P	94.53 _{0.013}	95.48 _{0.013}	245.4 _{1.273}	55 _{0.074}

- Brak dostrajania parametrów, nie wytłumaczyli skąd wzięli te konkretne, których użyli
- Mało wytłumaczony temat (czym jest pruning rate i jak go liczą?)

- Niektóre grafiki mało opisane, niewytłumaczone i nieczytelne, mało naukowe



Pytania do autorów:

- Jak wyglądałoby eksperymenty przy zastosowaniu bardziej zaawansowanych sieci?
- Jak przebiegłyby na innym zbiorze danych?
- Jak liczony jest pruning rate?

Ocena: Reject

Neutralny prompt do LLM:

ACO-Pruning for Deep Neural Networks: A Case Study in CNNs

GECCO '24 Companion · Dorighello, Delgado, Lüders, Pigatto · Universidade Tecnológica Federal do Paraná

Summary

This paper proposes ACO-P (Ant Colony Optimization Pruning), a dynamic pruning method for deep neural networks that replaces the secondary decision network used in prior work (RNN-P) with an ACO-based agent. Ants traverse a graph mirroring the CNN's layer structure and use a novel gradient-based heuristic called Global Average Gradient (GAG) to identify which convolutional kernels to prune during training. The method is validated on the MNIST dataset using a three-layer CNN, and compared against no-pruning, random pruning, and RNN-P baselines. ACO-P achieves a ~55% pruning

rate and ~95.5% test accuracy, outperforming RNN-P on pruning rate (~4×) and training speed (~3×), at the cost of a modest ~3.5% accuracy drop from the unpruned baseline.

Strengths

- **Novel combination.** First known use of ACO as a dynamic pruner that operates directly on CNN kernels *during* training — clearly differentiated from prior NAS-focused ACO uses.
- **GAG heuristic.** The Global Average Gradient mechanism is a principled, lightweight alternative to feature-map-based heuristics, grounded in the training signal rather than static weight magnitudes.
- **Strong efficiency gains.** ACO-P trains nearly as fast as the unpruned model and roughly 3× faster than RNN-P, while achieving a 4× higher pruning rate — a practically meaningful result.
- **Recovery mechanism.** Allowing pruned channels to be re-admitted to the network adds useful flexibility that static pruners lack, and is a thoughtful design choice.
- **Reproducibility.** Parameter settings, hardware details, and the number of independent runs (10) are clearly documented.

Weaknesses

- **Single dataset, simple benchmark.** All experiments use MNIST — a near-saturated benchmark with little variance between methods. Results on CIFAR-10, CIFAR-100, or ImageNet subsets are needed to support generalizability claims.
- **Accuracy gap unexplained.** The ~3.5% accuracy drop from the unpruned model is called "competitive" but is never ablated. It is unclear how much of this stems from GAG, the Ant System update rule, the recovery parameter, or the pruning frequency E_p .
- **No FLOPs or inference speedup reported.** Pruning rate is measured by channel count, not by actual computational savings (FLOPs, latency, memory footprint). The paper's own introduction warns that irregular sparsity may not translate to wall-clock savings — yet this is never measured.
- **Weak ACO variant.** The Ant System (AS) is the simplest ACO variant, outperformed by ACS or MMAS in most benchmarks. The choice is justified only by simplicity, and no sensitivity analysis on α , β , ρ is provided.
- **RNN-P comparison may be unfair.** RNN-P uses a secondary recurrent network that adds parameters — this inflates its "pruning rate" disadvantage. The comparison should be normalized by total parameter count or computational budget.
- **No statistical significance testing.** Standard deviations are reported but no t-tests or confidence intervals are used to assess whether differences between ACO-P and RNN-P accuracy are statistically significant.
- **Modern architectures absent.** The paper applies its method only to a hand-crafted toy CNN. Claims of generalizability to transformers and graph networks remain entirely speculative.

Questions for authors

1. How sensitive is ACO-P's accuracy and pruning rate to the pheromone parameters (α , β , ρ) and the recovery limit rec ? Has any hyperparameter sensitivity analysis been conducted?
2. What are the actual FLOPs and inference latency of the pruned vs. unpruned model? Does the structured (channel-level) pruning yield real hardware speedups?
3. The GAG heuristic assigns higher pruning probability to kernels with higher gradients — but high gradients typically indicate high learning signal. Can the authors justify why pruning high-gradient kernels is desirable rather than low-gradient (more redundant) ones?
4. How does ACO-P behave on datasets with higher inter-class complexity (e.g., CIFAR-10)? Does the pruning rate remain stable or collapse?
5. Would switching from the Ant System to a stronger ACO variant (e.g., ACS or MMAS) yield meaningful accuracy or pruning improvements?

Overall recommendation

4 / 10

Decision

Reject — significant revisions needed

Justification

ACO-P is a genuinely novel idea — applying ACO as a dynamic pruner during CNN training is, to the best of the reviewer's knowledge, original — and the GAG heuristic is a creative contribution. The paper is also well-written and clearly structured. However, the experimental scope is too narrow to support the claims made. MNIST is insufficient to benchmark a pruning method in 2024; the absence of FLOPs/latency measurements undermines the central efficiency claim; the accuracy drop is unablated; and the baseline comparison has methodological issues. The paper reads more like a workshop paper or preliminary proof-of-concept than a full contribution. A revised version with stronger datasets, ablation studies, and hardware measurements would be considerably more compelling.

Agresywny prompt:

Review: ACO-Pruning for Deep Neural Networks — REJECT

Score: 2/10 — Strong Reject

Summary of Problems

This paper proposes using Ant Colony Optimization for dynamic CNN pruning. After careful reading, I find it severely lacking on nearly every dimension expected of a NeurIPS submission. My concerns are extensive and fundamental.

Major Weaknesses

1. Experimental scope is embarrassingly narrow.

MNIST in 2024. MNIST. This dataset was a reasonable benchmark in 1998. It has ten classes, tiny 28×28 grayscale images, and is trivially solved by virtually any modern method. The authors acknowledge this limitation in passing but present it as merely a direction for future work. It is not a limitation — it is a fatal flaw. No serious pruning paper published in the last five years uses MNIST as its primary or sole benchmark. Where is CIFAR-10? CIFAR-100? ImageNet? Tiny-ImageNet? Without results on a dataset that actually stresses the model, every claim about pruning rates and accuracy preservation is meaningless.

2. The baseline comparison is rigged.

The authors compare against RNN-P (Lin et al., 2017 — seven years old) and a random pruner. That is it. Where is comparison against:

- Lottery Ticket Hypothesis (Frankle & Carlin, 2019)?
- L1-norm filter pruning (Li et al., 2017)?
- Network Slimming (Liu et al., 2017)?
- Any method published after 2018?

Choosing a 2017 baseline and a straw-man random pruner, then declaring victory, is not a fair evaluation. The RNN-P method was not even designed for MNIST — the authors are applying it outside its intended context and then criticizing its performance. This is cherry-picking dressed up as rigor.

3. The CNN architecture is trivially small.

Three convolutional layers with 16, 32, and 32 kernels. This is a toy model. Real pruning research addresses ResNet-50, VGG-16, MobileNet — architectures where pruning actually matters because the computational burden is real. The authors gesture at "scalability" in Table 3 with three tiny variants (CNN1, CNN2, CNN3), none of which approach realistic model sizes. The largest tested architecture (CNN3: 16×32×32×64) has approximately 100K parameters. ResNet-50 has 25 million. The claim that ACO-P "could be challenging for huge-size CNNs" applied to RNN-P while implying ACO-P scales better is pure speculation — they never actually test a huge CNN.

4. The pruning rate comparison is deeply misleading.

ACO-P achieves 55% pruning vs. RNN-P's 14%, and the authors present this as a massive win. But they bury a critical admission: RNN-P deliberately skips the first convolutional layer and adds an encoding layer. These are architectural choices that affect what "pruning rate" even means. You cannot directly compare pruning rates between methods with different prunable parameter sets and declare one

fourfold better. This requires a normalized comparison (e.g., FLOPs reduction, actual inference speedup measured empirically). Neither is provided.

5. No inference speedup is reported.

This is a pruning paper. The entire point of pruning is to make inference faster or cheaper. The authors report training time, but where is the inference latency comparison? Where is the FLOPs count before and after pruning? Where is memory footprint reduction? Reporting "pruning rate" as a percentage of channels removed is a proxy metric. The actual goal — faster inference — is never demonstrated. A 55% channel removal does not necessarily translate to anywhere near 55% speedup due to hardware parallelism, memory access patterns, and implementation overhead.

6. The GAG heuristic is inadequately justified and ablated.

The Global Average Gradient (GAG) is presented as a key contribution, yet the only ablation comparing it to the alternative (CNN weights + global average pooling) reports "below 80% accuracy" with no further details. What were the exact numbers? What other heuristics were tried? Why is gradient magnitude a good proxy for pruning suitability? High gradient components are actively learning — there is a reasonable argument that pruning high-gradient components is exactly backwards. The authors assert GAG is better but provide no theoretical justification and only a single, poorly described comparison point.

7. ACO hyperparameters are unjustified.

Table 1 lists $\alpha=0.7$, $\beta=0.3$, $\rho=0.1$, 20 ants, MAXit=50, R=5. No sensitivity analysis is provided for any of these. In a meta-heuristic paper, hyperparameter sensitivity is not optional — it is the core question of whether your method is robust or just tuned to work on this specific toy problem. How were these values chosen? Grid search? Random search? Expert intuition? No answer is given.

8. The "first time" claim is overclaimed.

The authors repeatedly assert this is "the first time a dynamic pruning strategy based on ACO has been proposed." This claim is not verified rigorously — it rests on the authors' literature survey, which is acknowledged to be incomplete. Academic novelty claims of this type require systematic search evidence, not author assertion. Furthermore, even if technically novel, "first to apply method X to problem Y" is not sufficient contribution without demonstrating that X is actually well-suited to Y, which brings us back to the inadequate experimental validation.

9. Statistical analysis is insufficient.

Ten runs with standard deviations is the minimum acceptable, and the authors provide this. But they report no significance testing whatsoever. The difference between ACO-P (95.48%) and RNN-P (93.42%) testing accuracy — is this statistically significant? Given that RNN-P has std 0.008 and ACO-P has std 0.013, these distributions likely overlap. Without t-tests or similar, these numbers prove nothing.

10. The recovery mechanism is hand-wavy.

Step 27 of Algorithm 1 describes recovering pruned channels if they haven't been re-selected within recxEp epochs. This is presented as a feature, but it fundamentally complicates analysis. What is the actual pruned network structure at inference time? Is the network dynamically changing structure during inference (true dynamic pruning) or does it settle to a fixed structure? The paper is

ambiguous. If inference uses a fixed structure, what structure? If truly dynamic, the pruning rate comparison with RNN-P is even more confounded.

11. Training time comparison is unfair.

ACO-P is compared favorably to RNN-P on training time (~245s vs ~683s). But ACO-P is applied to a model with 55% channel pruning while RNN-P operates on a model with only 14% pruning — of course a smaller model trains faster. You are not comparing equivalent computational workloads. This comparison is meaningless without controlling for model size.

Minor Weaknesses

- Equation 1 is just gradient descent. Presenting it as a contribution is padding.
- The paper is published at GECCO '24 Companion (a workshop proceedings), not a main track. The authors are submitting incremental workshop work to a top venue.
- Figure 3 is overly cluttered and the "Colony" to "Food Source" framing adds no scientific clarity.
- "Yelding" (sic) in the abstract suggests insufficient proofreading.

Verdict

This paper presents an interesting idea — using ACO for dynamic pruning — but validates it almost entirely on a toy problem from the 1990s, against outdated baselines, without the standard metrics the community actually cares about (inference FLOPs, latency, real-world datasets). Every major claim is either unsubstantiated or confounded by experimental design choices that favor the proposed method. The authors need to substantially redo their experiments on modern benchmarks and architectures before this work is ready for a top venue.

Recommendation: Strong Reject.